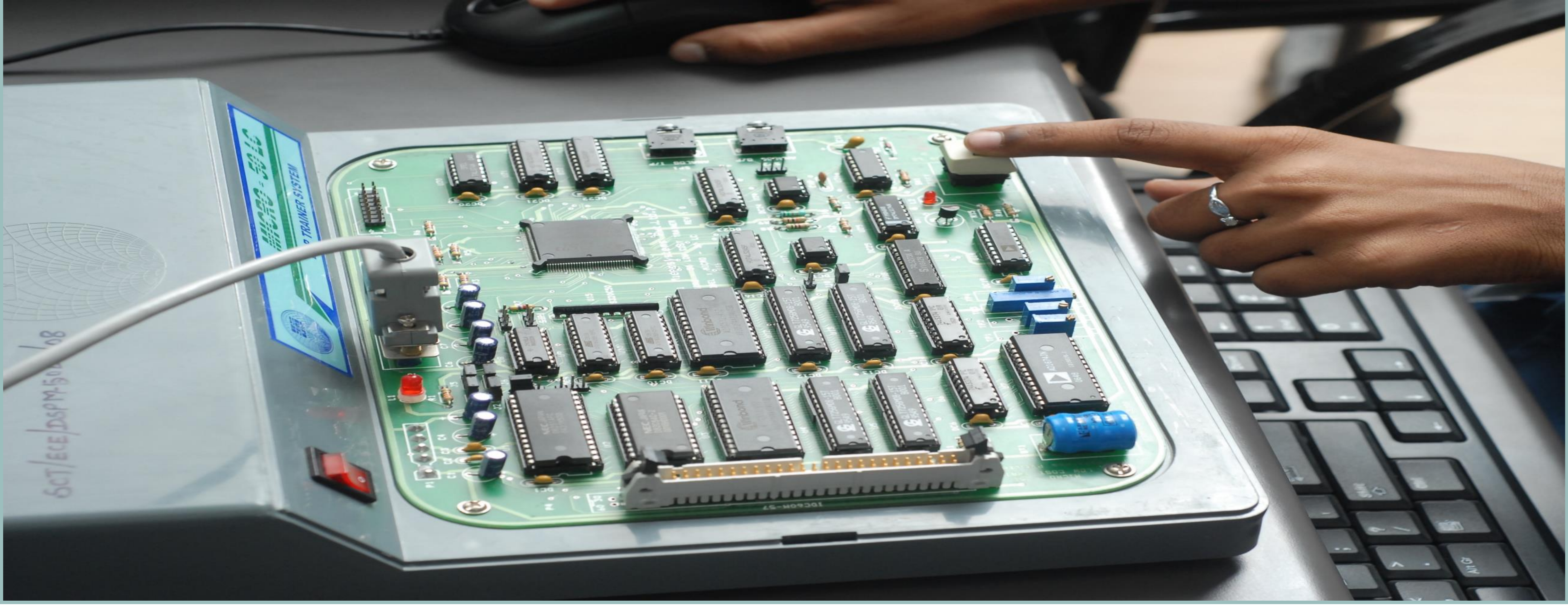


INTRODUCTION TO DIGITAL SIGNAL PROCESSING (2)

Notes By

Isi Edeoghon (PhD)

IMPORTANCE OF ANTI-ALIASING

When considering the bandwidth of the signal, we must also consider the frequency characteristics of any noise in the spectrum. Although the signal may have already been bandwidth limited by some other mechanism, e.g. by a phone system, there may be noise in the signal outside the signal bandwidth, and low-pass filtering may be required to keep the noise from aliasing.

NORMALIZATION

Normalization means to have a measure for a signal in the same, fixed, easy to use range such as [0,..,1] and this measure should not have physical units. For normalizing the frequency f to a value fn in range [0,...,1], the most obvious way is to divide f by the sampling frequency fs:

$$fn = f \,/\, fs \qquad (1)$$

Formula (1) is a number (no units) showing how many sampling periods with frequency fs are in the sampled signal f.

However, according to Nyquist-Shannon theorem, the sampling frequency is always at least two times the frequency f. Thus (1) is never larger than 1/2. To have *fn* in the range [0, 1], we multiply (1) with a factor 2:

$$fn = 2 * f \,/\, fs \qquad (2)$$

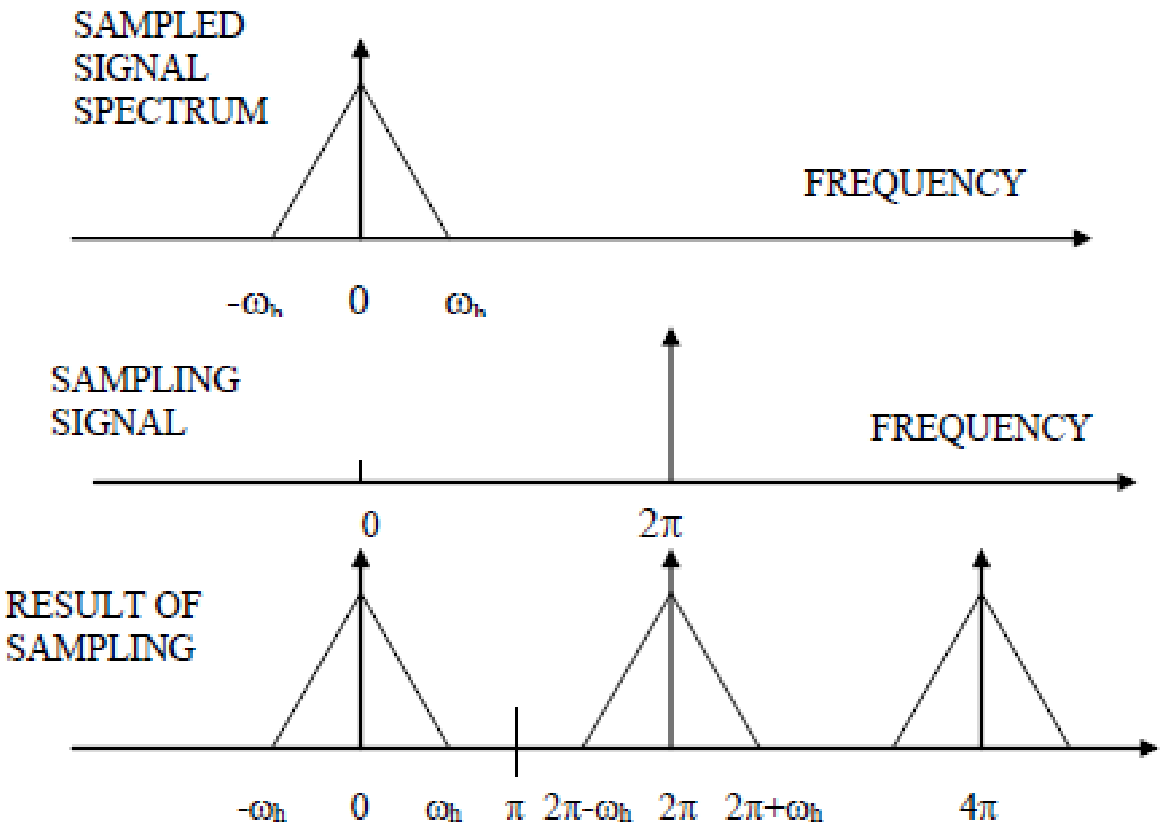
NORMALIZATION

Now if we replace fs to be the Nyquist frequency, $fs=2*f$ then we obtain *fn*=1. For higher sampling frequencies than Nyquist frequency, the value *fn* will be less than one since we divide 2*f by an increasingly large number, thus we obtain the range [0,..,1] for *fn* as we wanted. If one would not use the multiplier 2 in (2), then the range would be always [0,...,1/2] which is not so nice.

NORMALIZATION

When discussing sampled signals, it is customary to first normalize the frequencies involved. One approach is to normalize to the sampling frequency FS and use radian frequency instead of conventional cycles/second frequency. With these conventions, the sampling frequency becomes 1 in cycles/sec, or 2π in radian frequency. The symbol ω is used for frequency of the sampled variable, so that $\omega = 2\pi$ is the sampling frequency, and other frequency points are expressed in terms of that. Thus, the Nyquist frequency is π, and the spectrum of the signal will also be normalized to the sampling frequency. The spectrum of the sampled signal shown earlier would then be:

ILLUSTRATION OF NORMALIZED FREQUENCY



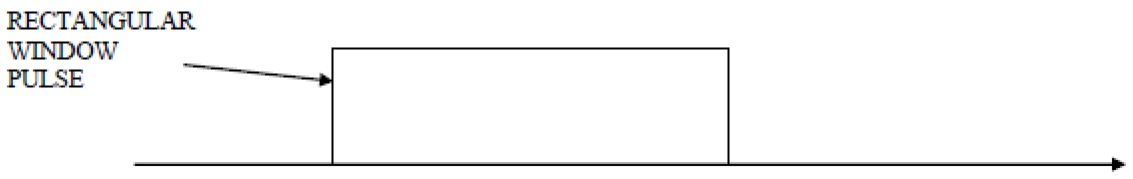
SAMPLING WINDOW

When a signal is sampled, the sampling operation generally starts abruptly - that is at some arbitrary point on the sampled waveform.

The portion of the signal that is sampled is generally a part of a much longer signal. The sampling starts abruptly at some point and stops at some other point.

The abrupt *start* and *stop* **introduce** frequency components into the sampled signal that did not exist in the original. The abrupt start and stop of sampling has the effect of multiplying the original analog signal with a rectangular pulse or "window" before sampling:

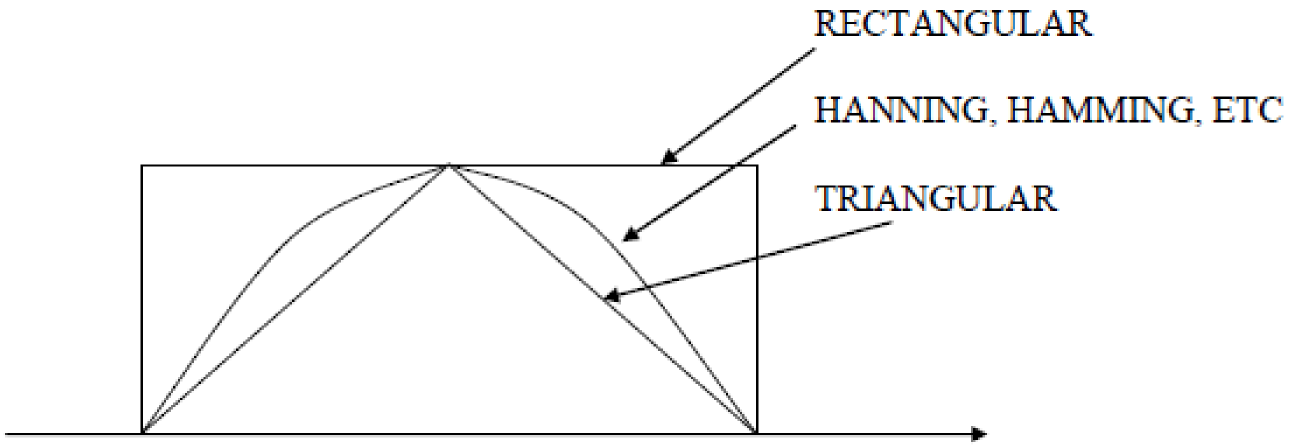
SAMPLING WINDOW



To reduce the effects of the abrupt transitions, it is customary to modify the shape of the original analog signal so that it approaches zero smoothly at the ends of the sample period, instead of abruptly.

This is done by multiplying by a **window function** which is selected to reduce the end effects while minimizing the resultant effect on the spectrum of the signal being sampled. The rectangular pulse is effectively the window used when no specific window is deliberately applied.

DSP STEP BY STEP



There are several standard windows whose characteristics in both the time domain and the frequency domain are well known. The general response of the popular ones are shown above.

The design of digital filters frequently uses windows, since one popular technique is to duplicate the impulse response of an analog IIR filter, and application of a window is necessary to limit the number of terms. In fact, one design technique is sometimes called the "windowing" approach.

EXERCISE

What is an IIR filter?

What are the differences between an IIR filter and an FIR filter?

Explain any of the windows in the preceding image?

TYPES OF ANALOG-TO-DIGITAL CONVERTERS (ADCS)

The Delta/Sigma ADC.

Dual-slope integrating ADC

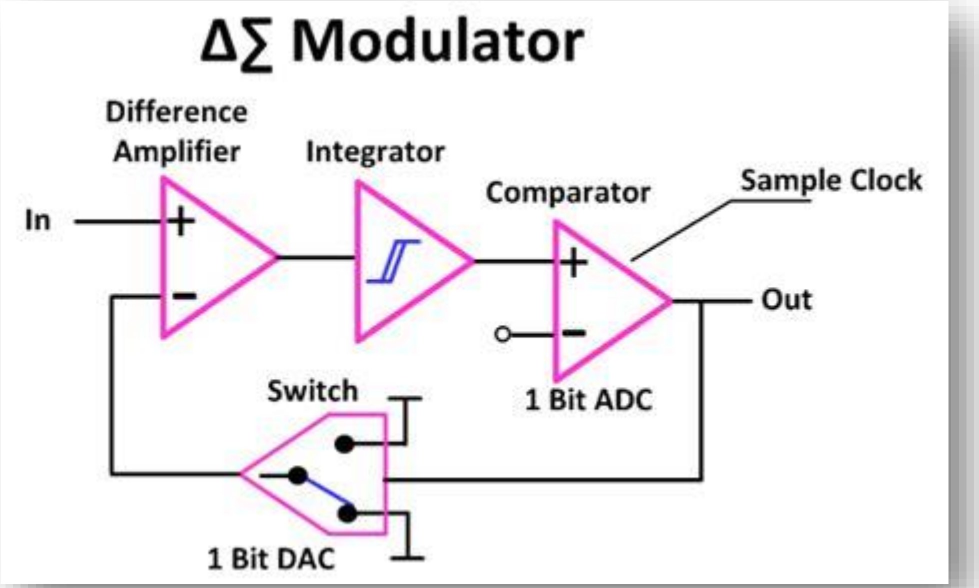
Charge-balance ADC

THE DELTA/SIGMA ADC

This type uses a simple one-bit converter to sample the input signal at a very high rate. The one-bit signal is then digitally filtered to produce a signal at a much lower rate, but with a higher resolution.

For example, the initial one-bit conversion could be at a rate of several MHz, which is then converted by a digital filter to a rate of tens of kHz, but with a resolution typically of 8 to 12 bits, but sometimes as high as 18 to 20 bits. The very high sampling rate means that sampling noise is well above the frequency range of the signal, so can be effectively filtered out by the digital filter.

THE DELTA/SIGMA ADC

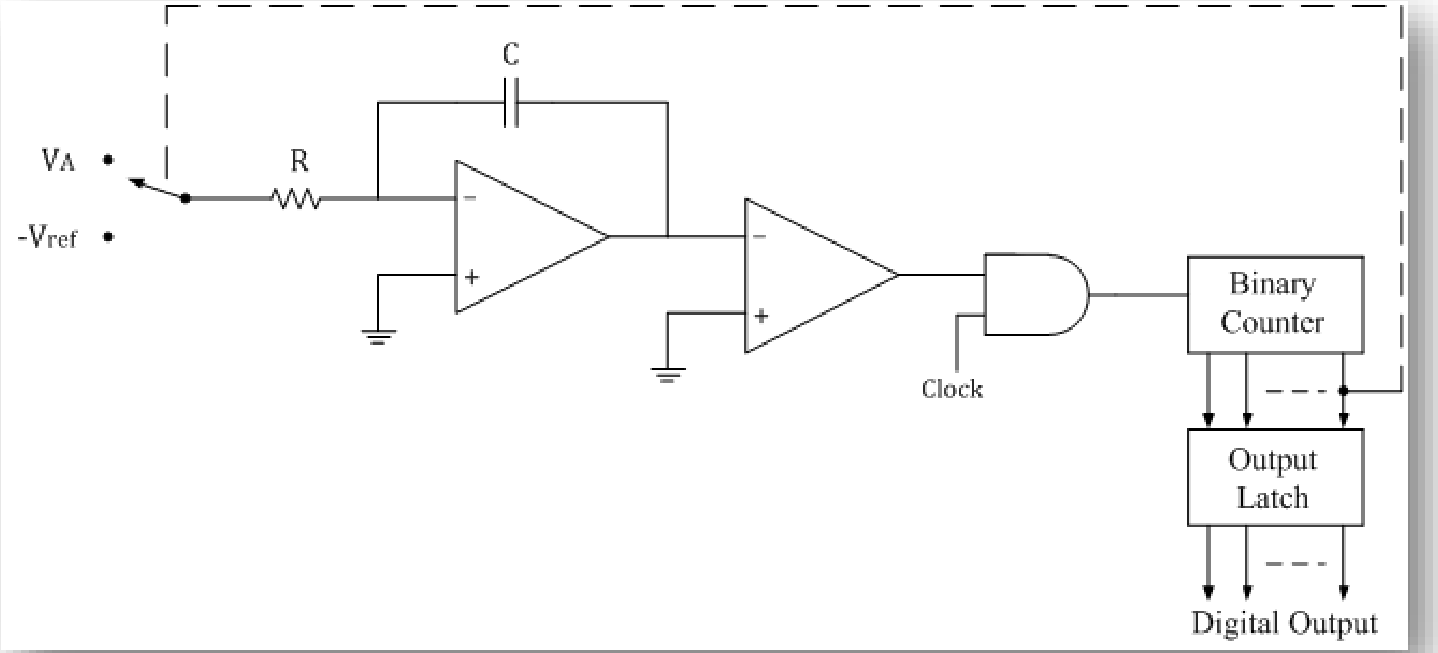


DUAL-SLOPE INTEGRATING ADC

The dual-slope integrating ADC counts the number of clock pulses required to discharge a capacitor at a rate proportional to the input signal, when the capacitor was initially charged to a level proportional to a reference voltage.

Thus, the number of counts represents the value of the input signal. Because the counting operation can be lengthy, and its time is not predictable, this type of ADC is more suitable for hand-held meters and other human interfaces than for DSP.

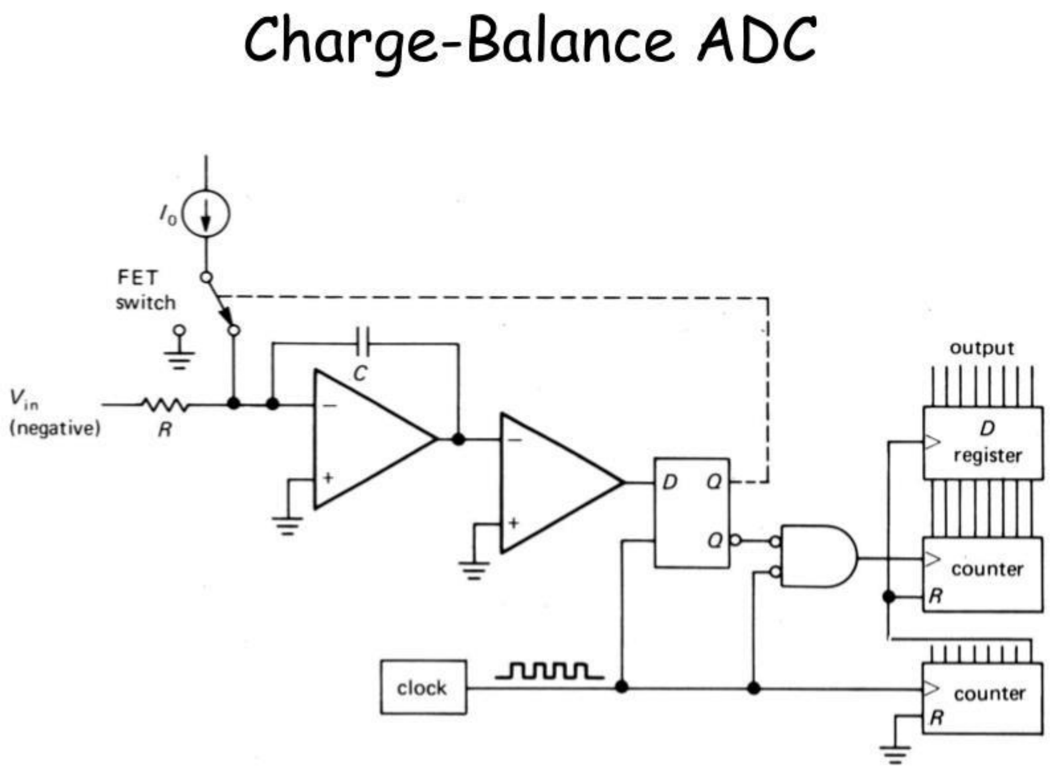
DUAL-SLOPE INTEGRATING ADC



CHARGE-BALANCE ADC

The charge-balance ADC operates in a similar fashion, except that it accumulates charge on a capacitor as it develops a counter value proportional to the value of the signal. It is also slow and not well suited for DSP. Note that both the dual-slope and the charge-balance ADCs require a conversion time proportional to the actual value (number of counts in the counter) of the input.

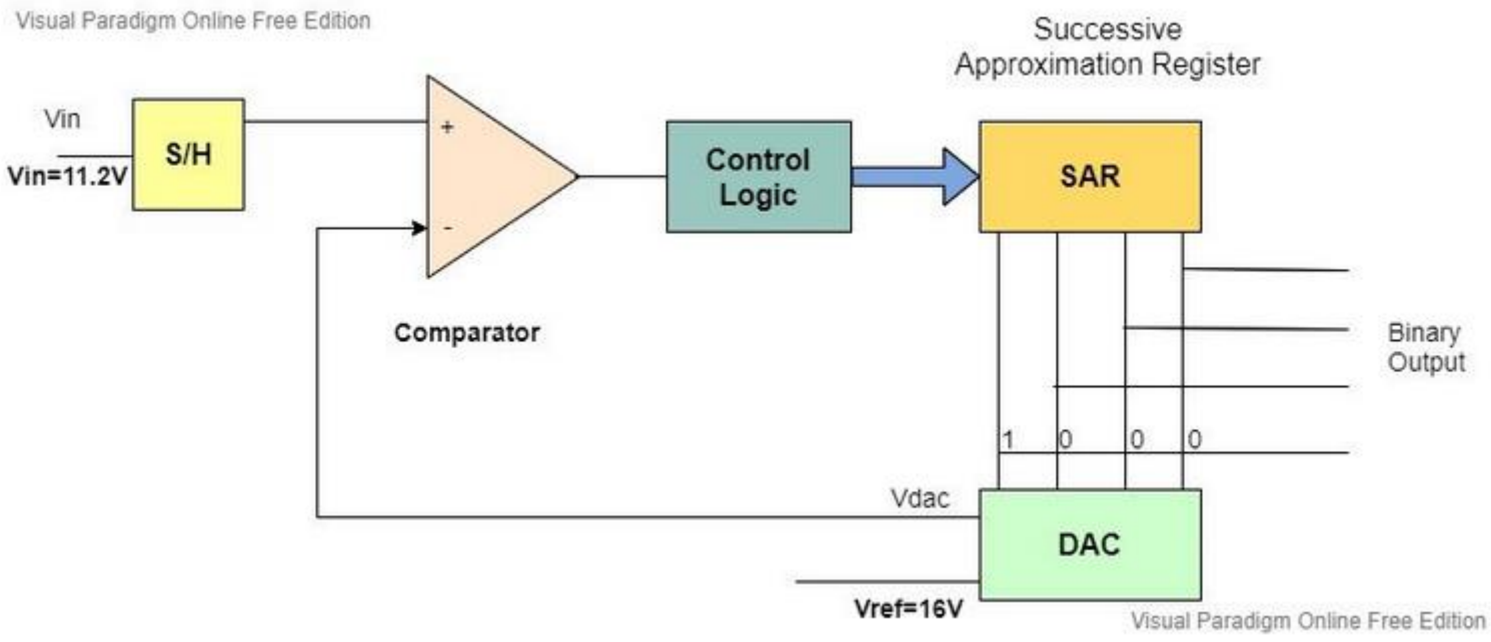
CHARGE-BALANCE ADC



SUCCESSIVE-APPROXIMATION ADC

The successive-approximation ADC uses a digital-to-analog converter (DAC) in a feedback loop to perform its conversion. It starts its operation by setting the output of the DAC to half scale, i.e. only the most significant bit is a ONE. It then compares the input to the DAC output. It resets the MSB if the input is lower, and keeps it if the input is higher. Then the input is compared with the MSB resulting from the previous step, added to the next most significant bit. Another decision is made, based on the size of the input relative to the DAC output.

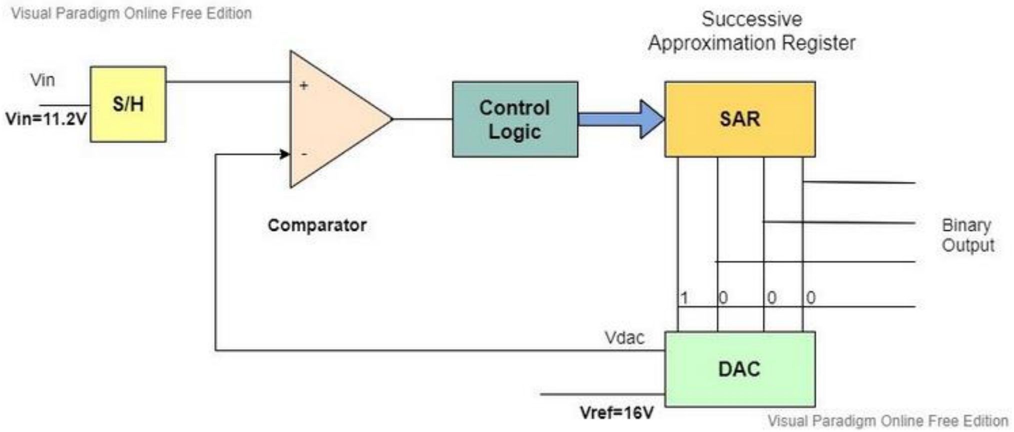
SUCCESSIVE-APPROXIMATION ADC



SUCCESSIVE-APPROXIMATION ADC

Consider the diagram preceding, a 4-bit ADC with a sampling rate i.e Vin to be 11.2 Volts. We take the comparator reference voltage as 16 Volts. Whenever the new transformation begins, the successive approximation register sets the most significant bit to 1 and all others to zero. As the register is followed by the DAC, the input to the DAC is 1000.

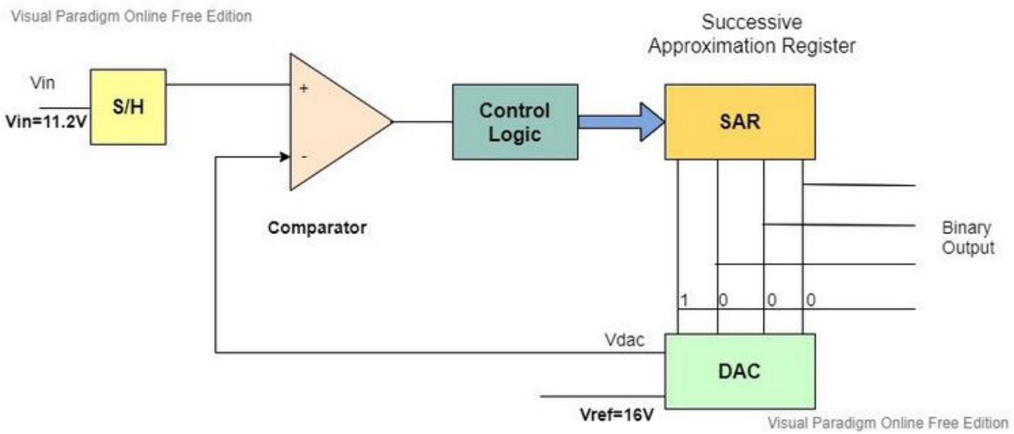
SUCCESSIVE-APPROXIMATION ADC



When Vdac < Vin

If the output voltage of DAC is less than the input voltage then the most significant bit remains intact and the next bit will be changed to 1 for new comparison.

SUCCESSIVE-APPROXIMATION ADC



When Vdac > Vin

On the other hand, if the output DAC voltage is greater than the input voltage then the MSB is transformed to zero but the next bit is pulled high for the new comparison.

SUCCESSIVE-APPROXIMATION ADC

Advantages

They output the binary representation serially.

They are highly accurate due to their high-resolution power.

Successive Approximation ADCs are reliable and power effective.

Disadvantages

As the resolution increases, the ADC slows down.

FINITE PRECISION PROBLEMS

In DSP work, there is always a concern because the precision of the data is limited. ADCs are based on a specific number of bits – anywhere from 7 or 8 for flash to as many as 20 for the delta-sigma.

Also, there is a loss of precision from the rounding that must be done to fit the results into the word size of the system.

FINITE PRECISION PROBLEMS

- ✓ QUANTIZATION NOISE
- ✓ LIMIT CYCLES

QUANTIZATION NOISE

✓ All of these effects mean that there is nearly always some difference between the number assigned to a sequence value by the limited-precision system, and what its real value would be with an infinite-precision system. The difference is generally uncorrelated from one sequence value to the next, so it is completely random. Thus, it can be treated as random noise, and its effect predicted on a statistical basis. In fact, the finite-precision effect can be analyzed in terms of signal-to-noise ratio, just like random noise in any system.

QUANTIZATION NOISE

✓Research and experimentation have shown that the effects of this "noise" depend on the word size (number of bits), the rounding scheme used in multi-stage calculations, and the binary code used (fixed vs. floating point). In general, 16 bits is adequate for most purposes, and decent results can be obtained down to 12 bits or fewer, depending on the applications. There is also the possibility for making some tradeoffs with other parameters. For example, in filter design, sometimes adding stages will allow lower precision in the numbers themselves.

LIMIT CYCLES

- ✓Another result of the finite precision available in DSP calculations is the production of limit cycles. These are constant amplitude output signals which appear at the output of a DSP process when the output is expected to approach zero. They occur because forcing the result of a computation to match one of the available output levels of a limited-resolution system - e.g. one of 256 for an 8-bit system or one of 65536 for a 16-bit system – is inherently a rounding or truncation process.
- ✓If the result of an operation is closer to one non-zero level, (say 0000 0001), than it is to zero (0000 0000), it will be rounded to 0000 0001 instead of just 0, and subsequent values may produce the same result, so that the output never reaches zero.

EXERCISE

Which ADC is the fastest and why?

How is Aliasing different from Quantization noise?

Where are ADC’s for DSP applied?

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